

In-Context Learning Assisted Prompt Optimization in LLMs for Classification Tasks

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DECLARATION

We, **Sree Vardhan Reddy (2022BCS0056)**, **Shaik Maleeha Afreen (2022BCS0212)**, **Nigade Vaishnav Dipakrao (2022BCD0045)**, and **Wakode Nikhil Niranjana (2022BCS0201)**, hereby declare that, this report entitled **“In-Context Learning Assisted Prompt Optimization in LLMs for Classification Tasks”** submitted to Indian Institute of Information Technology Kottayam towards partial requirement of **Bachelor of Technology in Computer Science and Engineering** is an original work carried out by me under the supervision of **Dr. Susheel Kumar Joshi** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold the academic ethics and honesty. Whenever an external information or statement or result is used then, that have been duly acknowledged and cited.

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April 2026

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ABSTRACT

The rapid growth of online news platforms has made misinformation easier to create and spread, leading to major social and political consequences. Although existing fake-news detection models often achieve high accuracy, they tend to function as opaque “black boxes” and perform inconsistently under noisy or shifted data conditions. To address this, the project introduces a Feature-Driven Optimization Framework (FDOF) that combines interpretable linguistic features with evolutionary optimization and modern LLM techniques. The framework constructs a broad set of features—such as readability, sentiment, and stylistic patterns—and applies a hybrid outlier-detection method using Mahalanobis distance and k-nearest-neighbour density. An Evolutionary Feature Selection Algorithm (EFSA) then selects an optimal feature subset by balancing model accuracy and simplicity. These optimized features are used to train transparent classical classifiers, which form the interpretable core of the system. To further enhance reasoning ability, the framework integrates a Retrieval-Augmented In-Context Learning (ICL) module that uses semantic retrieval to develop effective feature-informed prompts for a QLoRA-tuned Phi-2 model. Experiments demonstrate that FDOF achieves an accuracy of 84.11% with the optimized XGBoost model and up to 83.25% with the LLM using Dense Retrieval ICL. Overall, the framework delivers a more explainable, stable, and high-performing approach to fake-news detection by combining linguistic feature engineering with modern LLM advancements.

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Chapter 1

Introduction

1.1 Background and Motivation

The explosive growth of online platforms has radically accelerated how news spreads, but it has also amplified the reach and velocity of misinformation. Manual fact-checking is too slow and costly at scale, motivating automated, accurate, and *interpretable* fake news detection systems. Traditional machine learning (ML) and deep learning (DL) methods rely on large sets of lexical, syntactic, and semantic features, yet often treat all features as equally important, introducing noise and redundancy that weaken accuracy and explainability. Meanwhile, Large Language Models (LLMs) demonstrate strong language understanding but typically lack explicit feature attribution and depend on static, manually crafted prompts that are not tuned for task-specific objectives. These issues call for a more adaptive and transparent framework that *aligns feature relevance with model decision-making* while delivering robust detection performance.

1.2 Research Problem

We study binary fake news detection over labeled datasets of news headlines and bodies. Each sample is encoded as a semantic feature vector and assigned a binary label (*fake* vs. *true*). The central question is how to: (i) construct a comprehensive, content-based feature space; (ii) *select* a compact, informative subset to improve accuracy and interpretability; and (iii) *leverage* the refined features to guide LLM-based classification via prompt design and lightweight adaptation. The overarching goal is a pipeline that is accurate, explainable, and operationally practical.

1.3 Overview of the Proposed Framework

We adopt a two-stage pipeline, the **Fake Detection Optimization Framework (FDOF)**:

Stage I: Feature-driven Evolutionary Learning (FEL)

- **Feature Construction (220 features).** From the news headline and body, we extract a high-dimensional vector of *content* features spanning three axes: (a) *linguistic style* (e.g., POS usage, LIWC-derived style statistics, capitalization and stopword ratios), (b) *text complexity* (e.g., readability indices, type–token ratio, length and density statistics), and (c) *psychology* (e.g., LIWC categories, VADER sentiment, NRC emotion intensities, subjectivity/biased terms, punctuation signals).
- **Reliability Enhancement via Outlier Removal.** We identify and

remove anomalous samples by combining Mahalanobis Distance (MD) and K-Nearest Neighbors (KNN) detectors. A five-fold procedure evaluates outlier-removal choices; parameters for MD and KNN are tuned with Bayesian optimization to maximize average validation accuracy.

- **Evolutionary Feature Selection.** We apply a Genetic Algorithm (GA) to select a compact, informative feature subset. Binary chromosome encoding indicates inclusion/exclusion of features; tournament selection, crossover, and mutation evolve populations under a fitness function defined by cross-validated accuracy.
- **Classifier.** Using the GA-selected features, we train and compare multiple classifiers, including a Support Vector Machine (SVM), Random Forest (RF), and XGBoost, to identify the best-performing transparent model.

Stage II: Feature-Informed LLM Classification with Retrieval-Augmented ICL

- **Example Retrieval.** We leverage In-Context Learning (ICL) by retrieving semantic or lexical neighbors from the training corpus to serve as few-shot demonstrations.
- **Feature-Aware Prompting.** Prompt templates are dynamically constructed as *Feature-Informed Prompts*. We combine the retrieved shots with the article text and EFSA-selected "Indicators" (e.g., sentiment

scores and readability indices). Crucially, these indicators are injected into both the target query and the retrieved demonstrations.

- **QLoRA Adaptation.** The Large Language Model (LLM) is adapted via Quantized Low-Rank Adaptation (QLoRA) using 4-bit NormalFloat (NF4) quantization, aligning the model’s internal reasoning with the feature-informed prompts while minimizing computational overhead.

1.4 Contributions

1. A comprehensive content-based feature suite (220 features) covering linguistic style, complexity, and psychological semantics.
2. A reliability-enhanced training protocol that integrates MD and KNN outlier detection, optimized via a five-fold Bayesian tuning routine.
3. An Evolutionary Feature Selection Algorithm (EFSA) that yields compact, high-utility feature subsets.
4. A retrieval-augmented LLM pipeline utilizing QLoRA that outperforms standard fine-tuning by incorporating five distinct shot-selection strategies (Random, TF-IDF, BM25, SBERT, and Dense Retrieval).

1.5 Datasets and Experimental Setting (Summary)

We evaluate on PolitiFact (text + labels). Metrics include Accuracy, Precision, Recall, and F1, reported under five-fold protocols where appropriate. Empirically, (i) GA-based feature selection substantially boosts classical classifiers, with

strong gains observed for SVM, RF, and XGBoost; and (ii) for the LLM pipeline, integrating Retrieval-Augmented In-Context Learning (ICL) with QLoRA fine-tuning significantly outperforms zero-shot and standard fine-tuning baselines, with Dense Retrieval achieving the highest F1-score and bridging the performance gap with the best-performing classical ensembles.

1.6 Explainability and Transparency

FDOF advances explainability on two fronts: (i) the evolutionary selection process is auditable—each generation’s best subset and fitness trajectory can be recorded and visualized, revealing consistently chosen features. (ii) For the LLM pipeline, ”Feature-Informed Prompts” verbalize the exact metrics (e.g., readability, sentiment, top TF-IDF terms) directly in the context window. This prevents the LLM from operating as a pure ”black box” by grounding its reasoning explicitly on the engineered and statistical indicators discovered during Stage I.

1.7 Scope and Assumptions

Our study focuses on *text-only* fake news detection (headline + body). Social-context signals (e.g., user graphs) are out of scope for training the core pipeline. We assume labeled datasets with sufficiently clean text for standard preprocessing, and that parameter-efficient fine-tuning (QLoRA) alongside semantic retrieval indexes (Dense, SBERT) is feasible within typical academic hardware constraints.

Chapter 2

Theoretical Background

2.1 Text Classification Basics

Text classification maps a textual input $x \in \mathcal{X}$ to a label $y \in \mathcal{Y}$ by learning decision boundaries in a feature space derived from text. In supervised settings, we assume an i.i.d. sample $\{(x_i, y_i)\}_{i=1}^n$ and aim to learn a classifier $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ that generalizes beyond the training set. Common formulations include binary, multiclass, and multilabel classification. A typical pipeline comprises: (i) text normalization and tokenization, (ii) feature extraction or representation learning, (iii) model training, and (iv) evaluation on a held-out split to estimate generalization.

Classifiers span linear models (e.g., logistic regression, linear SVM), tree ensembles (RF/XGBoost), and deep neural architectures (CNN/RNN/Transformers). For modern LLM-based workflows, zero/few-shot inference can replace explicit training, shifting effort from parameter learning to *prompt design*. Regardless of the model family, reliable classification depends on clean data, well-specified

labels, and robust evaluation protocols that consider class imbalance and uncertainty.

2.2 Word and Document Representations

TF-IDF. Term frequency-inverse document frequency reweights terms by global rarity:

$$\text{tf}(t, d) = \frac{f(t, d)}{\sum_{t'} f(t', d)}, \quad \text{idf}(t, \mathcal{D}) = \log \frac{N + 1}{\text{df}(t) + 1} + 1,$$

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \text{idf}(t, \mathcal{D}).$$

TF-IDF attenuates ubiquitous tokens and highlights informative ones; cosine similarity in TF-IDF space is a strong baseline for clustering and retrieval [6].

2.3 Parameter-Efficient Fine-Tuning: QLoRA

To adapt Large Language Models under resource constraints, we employ Quantized Low-Rank Adaptation (QLoRA). This technique combines 4-bit quantization with Low-Rank Adapters (LoRA) to maintain model performance while significantly reducing memory overhead.

4-bit NormalFloat (NF4) Quantization

QLoRA uses an information-theoretically optimal 4-bit NormalFloat (NF4) format for the frozen weights $W \in \mathbb{R}^{d \times k}$. The quantization process maps weights to 16 discrete levels c_j derived from the quantiles of a standard normal

distribution $\mathcal{N}(0, 1)$. The quantized value q for a weight w is:

$$q = \arg \min_{j \in \{1, \dots, 16\}} \left| \frac{w}{\text{absmax}(W)} - c_j \right|$$

During the forward pass, the weights are dequantized to a higher precision (e.g., BF16) using the stored absolute maximum scale s :

$$\text{Dequantize}(s, q) = s \cdot c_q$$

Low-Rank Weight Updates and Forward Pass

Instead of updating the full weight matrix W , LoRA represents the update ΔW as the product of two low-rank matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$. The combined forward pass for a layer with input X is computed as:

$$Y = X \cdot \text{dequant}(W_{\text{NF4}}) + \frac{\alpha}{r}(XBA)$$

where r is the rank and α is a scaling factor. During fine-tuning, the model is optimized using a causal cross-entropy loss over the target sequence of length T :

$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^T \log P(x_i \mid x_{<i}; W_{\text{NF4}}, A, B)$$

The frozen base weights W_{NF4} do not receive gradients. Only the low-rank adapters are updated via stochastic gradient descent (or variants like AdamW) with learning rate η :

$$A \leftarrow A - \eta \cdot \nabla_A \mathcal{L}_{\text{CE}}, \quad B \leftarrow B - \eta \cdot \nabla_B \mathcal{L}_{\text{CE}}$$

This approach allows for significant memory savings while maintaining the expressive power required for task specialization.

2.4 Retrieval Strategies for In-Context Learning

In-Context Learning (ICL) effectiveness heavily depends on the relevance of the provided demonstrations. We formalize several paradigms for selecting these examples:

Stochastic and Lexical Retrieval

Lexical retrieval relies on exact keyword matching and statistical term distribution.

- **TF-IDF Similarity:** Documents are represented as sparse vectors of term weights. Retrieval is performed by maximizing the cosine similarity between the query vector q and the document vector d :

$$\text{sim}_{\text{TF-IDF}}(q, d) = \frac{\mathbf{v}_q \cdot \mathbf{v}_d}{\|\mathbf{v}_q\| \|\mathbf{v}_d\|}$$

where $\mathbf{v}_q$ and $\mathbf{v}_d$ are their respective TF-IDF representations (as defined earlier in Section 2.2).

- **Okapi BM25:** A robust probabilistic, non-linear extension of TF-IDF. Given a query Q and a document D , the score is calculated as:

$$\text{Score}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \cdot \frac{f(q, D) \cdot (k_1 + 1)}{f(q, D) + k_1 \cdot (1 - b + b \cdot \frac{|D|}{\text{avgl}})}$$

where $f(q, D)$ is the term frequency, $|D|$ is the document length, and $avgdl$ is the average document length across the corpus. Parameters k_1 and b control term frequency saturation and length normalization.

Semantic Retrieval: SBERT and Dense

Lexical methods often struggle with vocabulary mismatch (e.g., synonyms). Semantic retrieval resolves this by embedding text into a continuous high-dimensional vector space using transformer-based bi-encoders.

- **SBERT (Sentence-BERT):** Uses a symmetric bi-encoder architecture (e.g., `all-MiniLM`) optimized to produce embeddings where semantically similar sentences are close in the vector space. Similarity is strictly computed via **cosine similarity**:

$$\text{sim}_{\text{SBERT}}(q, d) = \frac{\phi(q) \cdot \phi(d)}{\|\phi(q)\| \|\phi(d)\|}$$

- **Dense Retrieval (Asymmetric):** Uses a bi-encoder optimized specifically for asymmetric search tasks, such as query-to-context retrieval (e.g., `multi-qa-MiniLM`). The similarity is computed directly via the **dot product** of their embeddings, allowing the model to capture asymmetrical relationships where the context d answers or supports the query q :

$$\text{sim}_{\text{Dense}}(q, d) = \phi(q)^T \psi(d)$$

2.5 Evaluation Metrics and Statistical Testing

Let TP_k, FP_k, FN_k denote counts for class k .

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \quad \text{Recall}_k = \frac{TP_k}{TP_k + FN_k}, \quad F1_k = \frac{2 \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}.$$

Averaging. Macro-F1 averages per-class F1 uniformly, highlighting minority classes; micro-F1 aggregates counts before computing the score, favoring frequent classes; weighted-F1 weights per-class F1 by support. Accuracy is informative for balanced data but can be misleading under skew.

Uncertainty. Report means $\pm$ standard deviation across reruns and optionally 95% bootstrap confidence intervals (resampling test instances) [5].

2.6 Ethical Considerations in Misinformation Detection

Labeling and Definitions. “Fake” and “real” labels often collapse complex editorial judgements; ambiguous or evolving stories risk mislabeling. Define label criteria transparently and document dataset provenance.

Explainability and Accountability. Systems should expose *why* a decision was made: feature attributions, example-based explanations, and indicator rationales support human audit. Feature-informed prompts that explicitly surface indicators improve transparency and can curb overreliance on superficial cues.

2.7 Summary

This chapter surveyed the core ingredients for text classification, spanning statistical representations like TF-IDF to the mechanics of modern Large Language Models. We specifically detailed the mathematical foundations of QLoRA adaptation encompassing NF4 quantization, double quantization, and low-rank gradient updates and formalized the lexical and semantic retrieval strategies required for In-Context Learning. These theoretical foundations support the methodology that follows, where we operationalize evolutionary feature selection and feature-informed LLM classification within a unified, reproducible pipeline.

Chapter 3

Literature Review

3.1 Existing Works

Taha et al. [7] survey genetic-algorithm (GA) feature selection from first principles (encoding, fitness design, selection/crossover/mutation) through hybrids that combine filter/embedded stages with GA wrappers. Their PRISMA-style review emphasizes GA’s strengths for high-dimensional search and local-optima escape, and highlights modern hybrids (e.g., GA-wrapper with neural models) that improve accuracy while pruning redundancy. The paper also surfaces practical issues: sensitivity to GA hyperparameters, computational cost from repeated fitness evaluations, and the need for standardized, domain-spanning benchmarks.

Che et al. [2] propose a multistage forecasting pipeline (CEEMDAN–GAFSF–DBiGRU–OLSSA) where GA-based wrapper selection (using BiLSTM as the base model) chooses informative inputs before deep modeling; CEEMDAN reduces volatility via decomposition; and an oppositional-learning sparrow search algorithm tunes

a stacked BiGRU forecaster. Across seasonal datasets, the pipeline reports improved error metrics, illustrating how wrapper-style GA selection aligned with model structure can yield measurable gains in nonstationary time-series settings.

Abdel-Salam et al. [1] propose HEST-HEO, an enhanced Human Evolutionary Algorithm for feature selection that combines Halton sequence initialization (diversity), an Enhanced Solution Quality operator, and a t -distribution disturbance to avoid local optima; they report gains on CEC’22 functions and 14 datasets, including higher accuracy with fewer features.

Yang et al. [8] introduce EMTFS, a multiform evolutionary multitask framework that solves feature selection as *both* a wrapper task (accuracy + sparsity) and a filter task (relevance/redundancy) in parallel using PSO, transferring “positive” knowledge between tasks via a two-channel strategy to improve accuracy while reducing evaluation cost.

Large Language Models and In-Context Learning

Beyond classical feature selection, the advent of Large Language Models (LLMs) has shifted classification paradigms towards prompt-based reasoning. Recent works [4] demonstrate that In-Context Learning (ICL)—where an LLM is provided with a few demonstration examples before the target query—can match or exceed fine-tuned models without parameter updates. However, standard ICL relies heavily on the semantic relevance of the retrieved shots. Techniques like Dense Passage Retrieval (DPR) and Sentence-BERT (SBERT) have been widely adopted to dynamically retrieve the most semantically similar demonstrations for a given query, creating a Retrieval-Augmented

Generation (RAG) pipeline for classification. Our work bridges these two domains: combining the rigor of evolutionary feature selection with the reasoning capabilities of retrieval-augmented LLMs.

Table 3.1: Summary of Selected Works in Feature Selection and Model Integration

Paper	Methodology & Inference	Limitations
Taha et al. (GA-FS)	Survey of GA encodings, fitness via CV, and hybrids; GA/hybrids reduce redundancy and boost generalization.	High eval cost; sensitive to hyperparameters.
Che et al. (GA + DBiGRU)	Wrapper FS (GA + BiLSTM), CEEMDAN decomposition, DBiGRU forecaster.	Domain-specific pipeline; complex stack.
Dong et al. (ICL & RAG)	Uses Dense Retrieval to find semantically similar demonstrations for LLM few-shot prompting.	Context window limits; sensitive to noisy retrieval.

Chapter 4

Dataset & Exploratory Analysis

This chapter documents the datasets used in this study, their schema and label taxonomy, our cleaning and normalization pipeline, the split strategy for fair evaluation, and an exploratory analysis (EDA) covering class balance, length and vocabulary statistics, and potential drift/bias across sources and time. The goal is to create a transparent, reproducible foundation for the modeling and optimization work in later chapters.

4.1 Dataset Description (schema, label taxonomy, sources)

Sources

We consider publicly curated fake/real news corpora commonly used for content-based detection. In our setup, each instance corresponds to a news item with (optionally) separate headline and body fields. Where available, we

also retain metadata such as outlet/source and publication date to support topic- and time-aware evaluation.

Schema

All datasets are normalized to a common tabular schema:

Field	Description
id	Stable unique identifier per instance (string or integer).
headline	Title/headline text (UTF-8). Empty if unavailable.
body	Main article text; minimally cleaned raw content.
label	Ground-truth class (binary: real/fake ; or mapped from fine-grained scales).

Split Policies

We consider three policies; the chosen one for this report is the random stratified split:

1. **Random stratified split:** preserves class ratios; simplest but sensitive to topic leakage.
2. **Chronological split:** training on earlier items, validation in the middle window, and testing on the most recent portion to emulate deployment. This reduces time leakage and better reflects real-world performance drift.

Visualization of Split Sizes

To illustrate the distribution of samples across the train, validation, and test sets, we include a bar plot summarizing the split proportions.



Figure 4.1: Train, validation, and test set sizes after applying the chosen split strategy.

K-folds and Model Selection

During prompt/model selection we use *train* for fitting and *validation* for tuning. For small datasets, we optionally employ k -fold cross-validation on training data to reduce variance; the test set is evaluated *once* after freezing the chosen configuration.

4.2 Class Balance & Imbalance Diagnostics

Let n_r and n_f denote counts of **real** and **fake**. We report:

$$\text{Imbalance Ratio (IR)} = \frac{\max(n_r, n_f)}{\min(n_r, n_f)}. \quad (4.1)$$

To better understand the class distribution, we include a bar chart showing the counts of fake vs. real items in the dataset.

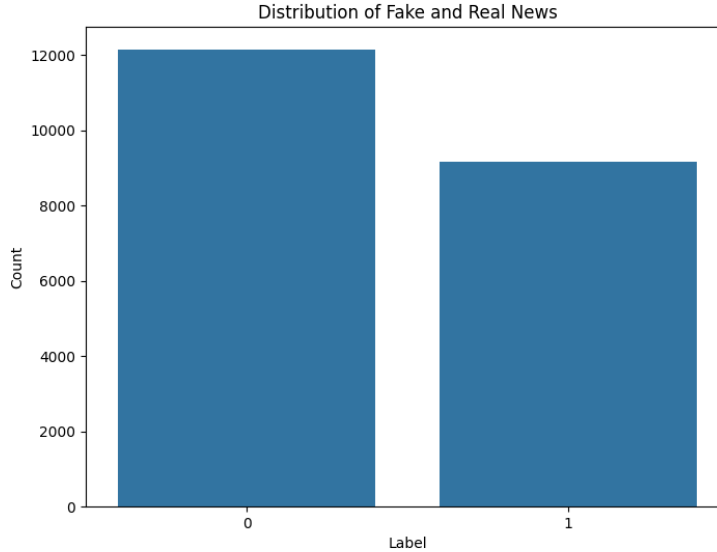


Figure 4.2: Distribution of fake (label 1) and real (label 0) news items in the dataset.

We include per-split bar charts of class counts and compute macro- vs. micro-averaged baselines to highlight skew sensitivity. For highly imbalanced data (e.g., $\text{IR} \geq 2$), we:

- prefer **accuracy** as the primary selection metric,

- audit per-class precision/recall and confusion matrices,
- consider class-aware sampling only within training (never in validation/test),
- avoid threshold tuning on test data (validation-only).

4.3 Summary of EDA Findings

- Basic data cleaning is performed, with further normalization handled during feature extraction.
- A random stratified split is adopted to preserve class ratios across the train, validation, and test sets.

These EDA outcomes directly inform the modeling choices in Chapter 5 and the selection/evaluation protocols in Chapter 9.

Chapter 5

System Design & Architecture

This chapter details the system architecture for fake news detection that combines robust, classical content modeling with feature-informed Large Language Model (LLM) classification. The design is organized as a two-stage pipeline: (i) **Feature-driven Evolutionary Learning (FEL)** for content feature construction, reliability enhancement, and evolutionary feature selection; and (ii) **Feature-informed LLM classification** that injects selected indicators into constrained prompts (with optional lightweight adaptation). The focus is on clear module responsibilities, well-defined interfaces, and deterministic control flow for reproducible evaluation.

5.1 End-to-End Pipeline Overview

Input. Cleaned, split datasets from Chapter 4, each instance providing `headline`, `body`, and a binary `label`.

Stage I: Feature-driven Evolutionary Learning (FEL).

1. **Content Feature Construction.** Compute a comprehensive vector of linguistic style, text complexity, and psychology/affect features per article (headline+body).
2. **Reliability Enhancement.** Screen anomalies with a hybrid outlier routine (e.g., Mahalanobis distance + KNN density) to stabilize downstream selection and training.
3. **Evolutionary Feature Selection.** Run a genetic algorithm (GA) to choose a compact, informative subset; fitness is cross-validated performance of a transparent classifier.
4. **Transparent Baseline Classifier.** Train a linear SVM (or similar) on the evolved subset; retain margin/support-vector diagnostics for auditability.

Stage II: Feature-Informed LLM Classification with Retrieval-Augmented ICL.

1. **Example Retrieval.** For each test instance, a retrieval module finds the $k = 7$ most relevant samples from the training set using lexical or semantic similarity.
2. **Feature-Informed Prompting.** A structured prompt is constructed that explicitly pairs the text with the EFSA-selected "Indicators." Crucially, these feature-informed prompts are applied to both the k retrieved demonstration examples and the target query, guiding the LLM's attention to the engineered signals.

3. **QLoRA Adaptation.** The LLM is fine-tuned using 4-bit quantization and low-rank adapters to specialize its reasoning for the indicator-heavy prompt schema.
4. **Constrained Inference.** At test time, a logits processor restricts the model to outputting only the single-digit classification labels.

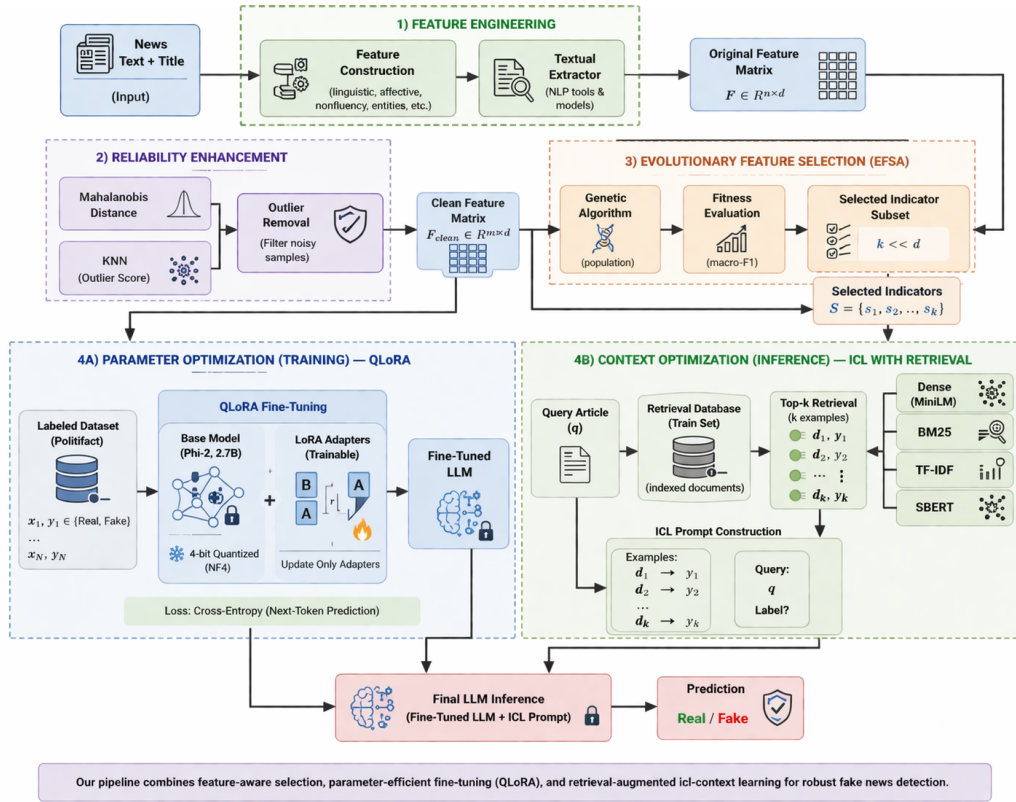


Figure 5.1: Overall architecture of the Fake Detection Optimization Framework (FDOF), illustrating the two-stage pipeline: Feature-Driven Evolutionary Learning and Feature-Informed LLM Classification.

5.2 Retrieval Module

The retrieval module is responsible for the "In-Context" portion of the pipeline. It maintains a structured index of the training data and provides five distinct interfaces for example selection:

- **TF-IDF Similarity:** Employs sparse vectors and cosine similarity to find examples with high lexical term overlap.
- **Okapi BM25:** Ranks examples based on a probabilistic model that accounts for term frequency saturation and document length.
- **Dense Retrieval:** Projects the query into a continuous latent space using an asymmetric bi-encoder (e.g., `multi-qa-MiniLM`) optimized for dot-product similarity.
- **SBERT Retrieval:** Uses a symmetric Sentence-BERT encoder (e.g., `all-MiniLM`) optimized for semantic sentence embeddings and cosine similarity.

The module outputs a formatted block of k demonstrations, each following the *News + Indicators + Label* format.

Outputs. Final metrics (accuracy, macro-/micro-F1, etc.), per-class diagnostics, confusion matrices, and human-auditable artifacts (selected features and prompts).

5.3 Control Flow and Orchestration (Conceptual)

The system follows a deterministic, stage-gated control flow:

1. **Prepare.** Load splits, normalize text, and verify de-duplication constraints.
2. **Reliability Pass.** Apply outlier masks; maintain a census of kept/removed instances.
3. **Stage I.** Compute features $\rightarrow$ run GA selection $\rightarrow$ train/evaluate SVM/XGBoost/Random Forest with cross-validated model selection based on accuracy.
4. **Stage II.** Retrieve demonstrations via ICL $\rightarrow$ Format feature-informed prompts $\rightarrow$ Supervised fine-tuning via QLoRA $\rightarrow$ Validation scoring.

All data-driven choices (thresholds, feature subsets, retrieval indices, prompts) are derived *only* from train/validation to avoid test leakage.

5.4 Data Loader & Preprocessing Module

Interfaces.

- `load_splits()`: returns train/val/test iterables with schema fields.
- `normalize(doc)`: applies order-sensitive normalization; returns transformed text plus a sidecar log (length deltas, masked counts).
- `apply_outlier_mask(X)`: excludes/quarantines instances; returns filtered indices.

5.5 Feature Extraction Layer

Feature families and rationale.

- **Linguistic Style.** POS distributions, case/caps emphasis, stopword ratios, headline/body asymmetry; captures stylistic regularities often correlated with sensationalism.
- **Text Complexity.** Readability indices, length/density, lexical diversity; approximates cognitive load and editorial polish.

Outputs. (i) A dense feature matrix for selection and classical modeling; (ii) a direct formatting of selected feature names and their numeric values for use in the prompting layer.

5.6 Model Training & Evaluation Module

Reliability & Selection. After the reliability pass, the GA explores binary masks over the feature set. Tournament selection, crossover/mutation, and patience-based early stopping manage efficiency. Fitness uses stratified k -fold validation on accuracy.

Transparent Baseline. A linear SVM is trained on the selected subset; decision margins and support vectors are retained for post hoc analysis.

Feature-informed LLM with ICL. A retrieval module finds the k most relevant examples from the training split. These demonstrations, alongside the article text and the EFSA-selected indicators, are assembled into a structured prompt with a *fixed answer set*. Decoding is strictly constrained via a Logits Processor to reduce off-manifold outputs. The LLM’s reasoning is then specialized using 4-bit QLoRA fine-tuning on this schema.

Testing Protocol. The best configuration (features, indicators, prompt, decoding) is frozen after validation. A single test run produces primary

metrics, per-class precision/recall/F1, and confusion matrices.

5.7 Remarks

The architecture emphasizes *robustness* (outlier screening), *parsimony* (evolutionary selection), *transparency* (verbalized indicators and SVM margins), and *budget-awareness* (frozen selection, constrained decoding). By decoupling feature engineering from prompting, the system leverages complementary strengths: interpretable content cues to focus the LLM’s attention and scalable classical baselines that provide reliable lower bounds on performance.

Chapter 6

Baseline Models

Scope note. This chapter establishes *training-based* baselines only. We *exclude* zero/few-shot prompting and broader in-context learning from the baseline suite, as these paradigms form the core of the proposed Stage II framework detailed in Chapter 7. The feature-informed LLM approach is evaluated in later chapters against the classical and SFT-only baselines defined here.

6.1 Baseline Model: Linear SVM with Engineered Features

Representations

The primary baseline relies on a comprehensive 220-dimensional feature vector computed from the cleaned text (Chapter 4). This vector combines two families of features:

- **TF-IDF Features:** A 200-dimensional vector representing token importance, computed using a TF-IDF vectorizer. Key parameters like n-gram range and document frequency cutoffs are tuned.
- **Engineered Features:** A 20-dimensional vector of handcrafted features capturing linguistic style (e.g., word counts, punctuation ratios), text complexity (e.g., readability scores), and psychological valence (e.g., VADER sentiment scores).

The headline and body are concatenated before features are extracted. This combined representation provides a rich, interpretable input for the classifier.

Linear SVM

A strong, sparse baseline with hinge loss is used for classification [3]:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_i \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)).$$

We prefer a linear kernel for the high-dimensional feature space. The regularization parameter C is tuned on the validation set. The primary metric for model selection is cross-validated accuracy.

6.2 Tree-based Ensemble Baselines

To complement the linear SVM, we evaluate two powerful tree-based ensemble models known for their strong performance on tabular data.

Random Forest

Random Forest is an ensemble learning method that operates by constructing a multitude of decision trees at training time. For a classification task, the final decision is made by a majority vote of the individual trees. It uses bagging and feature randomness to create uncorrelated trees, which helps to control overfitting and improve predictive accuracy.

For a classification task, the final prediction of a Random Forest model is the mode of the predictions from all the individual decision trees. If the forest consists of B trees $\{T_1, T_2, \dots, T_B\}$, the prediction for a new input $\mathbf{x}$ is:

$$\hat{y}_{rf} = \operatorname{argmax}_{c \in \mathcal{Y}} \sum_{b=1}^B I(\hat{y}_{T_b}(\mathbf{x}) = c)$$

where $\hat{y}_{T_b}(\mathbf{x})$ is the prediction of the b -th tree and $I(\cdot)$ is the indicator function. This represents a majority vote over all trees.

XGBoost

XGBoost (Extreme Gradient Boosting) is an optimized, distributed gradient boosting library. It is a highly effective and widely used implementation of gradient boosted decision trees. The model is built sequentially, with each new tree correcting the errors of the previous ones. Its objective function includes a regularization term to prevent overfitting:

$$\text{Obj}(\Theta) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where l is the loss function, and Ω is the regularization term that penalizes the complexity of the model’s trees.

6.3 Causal Language Model Baseline

We fine-tune a pretrained causal language model (`sshleifer/tiny-gpt2`) using a feature-informed prompting strategy to serve as a generative classification baseline:

- **Feature-informed Prompt:** For each sample, a prompt is constructed that includes the top TF-IDF features and the engineered feature values. This provides the model with explicit signals derived from the feature engineering stage.
- **Task Formulation:** The model is trained to output a single digit ('1' for fake, '0' for real) by predicting the next token. The optimization minimizes the standard auto-regressive Cross-Entropy loss over the sequence length T :

$$\mathcal{L}_{\text{CLM}} = - \sum_{t=1}^T \log P(w_t \mid w_1, \dots, w_{t-1}; \Theta)$$

where Θ represents the model parameters.

- **Training:** The model is adapted using standard supervised fine-tuning. Key hyperparameters like learning rate, batch size, and number of epochs are tuned on the validation set.

This baseline tests the effectiveness of injecting structured, engineered features into the reasoning process of a generative language model.

6.4 Implementation Notes & Hyperparameter Grids

Common Protocol

All baselines are trained on `train`, selected on `val`, and reported once on `test` with accuracy as the primary metric. Hyperparameters are tuned *only* on validation. For the SVM, stratified k -fold CV within `train` is used to stabilize selection.

Hyperparameter Ranges (for selection on validation)

- **Linear SVM:** $C \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$; `class_weight` $\in \{\text{None}, \text{balanced}\}$.
- **Random Forest:** `n_estimators` $\in \{100, 200, 300\}$; `max_depth` $\in \{10, 20, \text{None}\}$; `min_samples_leaf` $\in \{1, 2, 4\}$.
- **XGBoost:** `n_estimators` $\in \{100, 200, 300\}$; `max_depth` $\in \{3, 5, 7\}$; `learning_rate` $\in \{0.05, 0.1, 0.2\}$; `subsample` $\in \{0.8, 1.0\}$.

Imbalance handling. While accuracy is the primary metric, we report per-class precision/recall to monitor performance on the minority class. Class weights are considered for the SVM. No rebalancing is performed on validation/test sets.

6.5 Baseline Results & Early Insights

Table 6.1 presents the initial performance comparison between the traditional Linear SVM and the causal language model (`tiny-gpt2`) using feature-informed prompts.

Table 6.1: Baseline Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Linear SVM	0.8158	0.7636	0.8316	0.7961
Tiny-GPT2 (Fine-tuned)	0.7912	0.8312	0.6490	0.7289

6.5.1 Per-Class Performance

A detailed breakdown for each class (Table 6.2) shows that while both models perform well on Real News (Class 0), the Linear SVM demonstrates better recall on Fake News (Class 1) compared to the Tiny-GPT2 model, which appears to be more conservative in identifying misinformation in its base fine-tuned state.

Table 6.2: Per-Class Metrics for Baseline Models

Model	Class	Precision	Recall	F1-Score
Linear SVM	Real (0)	0.8542	0.8122	0.8327
	Fake (1)	0.7636	0.8316	0.7961
Tiny-GPT2	Real (0)	0.7707	0.8995	0.8302
	Fake (1)	0.8312	0.6490	0.7289

6.5.2 Early Insights

- **SVM Robustness.** The Linear SVM, utilizing the full 220-dimensional feature set, serves as an exceptionally strong baseline, often outperforming the smaller language model in raw detection metrics.
- **LLM Bias.** The Tiny-GPT2 model shows a higher precision for the Fake class but lower recall, suggesting that when it does label an item as fake, it is likely correct, but it misses a significant portion of misinformation.

6.6 Summary

This chapter defined the training-based baselines for our study: a Linear SVM using a rich, combined feature set, and a fine-tuned Causal Language Model guided by feature-informed prompts. These baselines provide reliable anchors for comparison in subsequent chapters, where the full FDOF pipeline is evaluated.

Chapter 7

Methodology: Feature Optimization and LLM Adaptation

This chapter formalizes the two-stage optimization approach used in our framework. We first evolve a compact, informative *indicator* set via Evolutionary Feature Selection (EFSA). We then leverage these indicators to guide a Large Language Model (LLM) through a combination of Parameter-Efficient Fine-Tuning (PEFT) and Retrieval-Augmented In-Context Learning (ICL).

7.1 Problem Formulation

Let $S = \{(t_i, y_i)\}_{i=1}^m$ be a dataset of news items where t_i represents the text and $y_i \in \{0, 1\}$ represents the label (1 for fake, 0 for real). Each text t is mapped to a high-dimensional feature space $F_t \in \mathbb{R}^d$, where $d = 220$.

Our objective is to find an optimal binary mask $g \in \{0, 1\}^d$ and an adapter configuration θ that maximizes the classification accuracy:

$$\max_{g, \theta} \frac{1}{m} \sum_{i=1}^m \mathbb{I}(\text{LLM}(t_i, \text{Prompt}(F_i \odot g); \theta) = y_i)$$

where $\odot$ denotes the element-wise product (selection), and Prompt represents the transformation of selected features into a textual "Indicator" block.

7.2 Stage I: Feature-Driven Evolutionary Learning (FEL)

7.2.1 Feature Construction and Normalization

Before optimization, each document is transformed into a dense 220-dimensional feature vector. This comprises 200 TF-IDF statistical features and 20 explicitly engineered features capturing linguistic style, psychology, and complexity.

For linguistic style, we compute metrics such as the Type-Token Ratio (TTR) to measure lexical diversity:

$$\text{TTR} = \frac{\text{Number of Unique Words}}{\text{Total Number of Words}}$$

To capture the cognitive load required to process a news item, we calculate readability indices such as the Flesch Reading Ease (FRE), the Flesch-Kincaid Grade Level (FKGL), and the Gunning Fog Index (GFI):

$$\text{FRE} = 206.835 - 1.015 \left(\frac{\text{total words}}{\text{total sentences}} \right) - 84.6 \left(\frac{\text{total syllables}}{\text{total words}} \right)$$

$$\text{FKGL} = 0.39 \left(\frac{\text{total words}}{\text{total sentences}} \right) + 11.8 \left(\frac{\text{total syllables}}{\text{total words}} \right) - 15.59$$

$$\text{GFI} = 0.4 \left[\left(\frac{\text{total words}}{\text{total sentences}} \right) + 100 \left(\frac{\text{complex words}}{\text{total words}} \right) \right]$$

where "complex words" are defined heuristically as words containing three or more syllables. All extracted numerical features are standard-scaled (Z-score normalized) before selection:

$$z = \frac{x - \mu}{\sigma}$$

7.2.2 Reliability Enhancement (Outlier Removal)

Before feature selection, we stabilize the training set by removing anomalies in the feature space $\mathbb{R}^{220}$. We employ a hybrid strategy combining:

1. **Mahalanobis Distance (MD):** Captures multivariate outliers by accounting for feature correlations:

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{S}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$$

We use the Ledoit-Wolf shrinkage estimator for the covariance matrix $\mathbf{S}$ to ensure stability in high-dimensional settings.

2. **K-Nearest Neighbors (KNN):** Identifies local density anomalies. For a given sample $\mathbf{x}_i$, its KNN outlier score s_i is computed as the average Euclidean distance to its k nearest neighbors $N_j(\mathbf{x}_i)$:

$$s_i = \frac{1}{k} \sum_{j=1}^k \|\mathbf{x}_i - N_j(\mathbf{x}_i)\|_2$$

A sample is flagged as an outlier if s_i exceeds a statistically determined threshold $\mu + e_{knn}\sigma$.

We tune the thresholds (e_{md}, e_{knn}) and neighborhood size (k) using Bayesian Optimization to maximize downstream cross-validation accuracy.

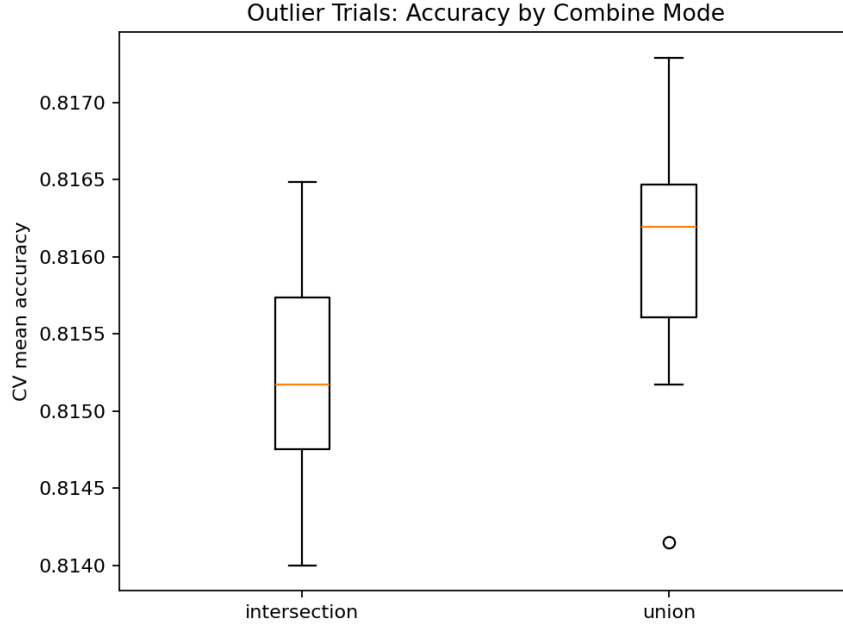


Figure 7.1: Accuracy comparison between intersection-based and union-based outlier removal strategies.

7.2.3 Evolutionary Feature Selection Algorithm (EFSA)

To prune the 220 features to a compact "Indicator" set, we use a Genetic Algorithm (GA). The population $P(t)$ at generation t consists of binary chromosomes $g \in \{0, 1\}^{220}$.

- **Fitness Function:** The fitness $f(g)$ of a chromosome is the mean

accuracy of a Linear SVM evaluated across a 5-fold stratified cross-validation on the training data X_{train} :

$$f(g) = \frac{1}{5} \sum_{k=1}^5 \text{Accuracy}(\text{SVM}(X_{train,-k} \odot g), X_{val,k} \odot g)$$

where $X_{val,k}$ is the validation split of fold k .

- **Selection:** We use Tournament Selection ($k = 3$), where 3 individuals are chosen at random and the one with the highest $f(g)$ is selected for reproduction.
- **Crossover and Mutation:** We apply uniform crossover ($p_c = 0.9$) to generate offspring c_1, c_2 from parents p_1, p_2 , followed by bit-flip mutation ($p_m = 0.01$).
- **Patience:** Early stopping is triggered if $\max_g f(g)$ does not improve for 6 generations.

7.3 Stage II: LLM Adaptation via QLoRA

We adapt a pre-trained LLM (e.g., `microsoft/phi-2`) using Quantized Low-Rank Adaptation (QLoRA) to align its reasoning with our selected features.

7.3.1 4-bit Quantization and NF4

To manage computational constraints, the base model weights W are quantized to 4-bit using the NormalFloat (NF4) data type. This format is information-theoretically optimal for normally distributed weights. We also employ

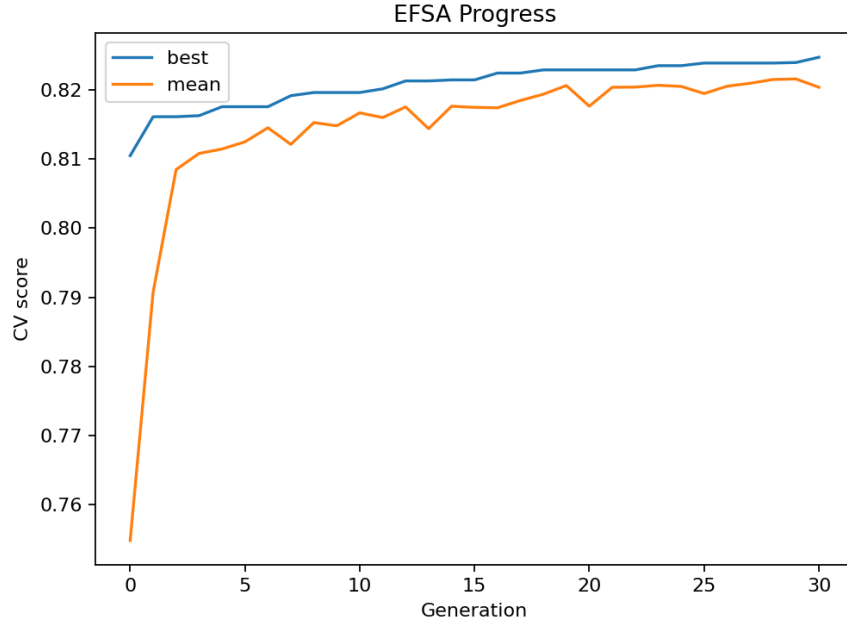


Figure 7.2: EFSA convergence: Best and mean validation accuracy across generations.

Double Quantization (DQ), which quantizes the first-level quantization constants s_1 into 8-bit floats with a second-level scale s_2 , reducing memory footprint without sacrificing precision:

$$q = \arg \min_j \left| \frac{w}{\text{absmax}(W)} - c_j \right|, \quad s_1 = s_2 \cdot c_{q_2}$$

where c_j are the quantiles of a standard normal distribution.

7.3.2 Low-Rank Update and Training Objective

Trainable low-rank matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$ are added to the frozen quantized weights. The forward pass is defined as:

$$Y = X \cdot \text{dequant}(W_{\text{NF4}}) + \frac{\alpha}{r}(XBA)$$

where r is the rank and α is a scaling factor. We target the `q_proj`, `k_proj`, `v_proj`, and `dense` layers. The model is optimized using a causal cross-entropy loss $\mathcal{L}_{\text{CE}}$ over the sequence:

$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^T \log P(x_i \mid x_{<i}; W_{\text{NF4}}, A, B)$$

During the backward pass, gradients are computed only for the adapters, which are updated via:

$$A \leftarrow A - \eta \nabla_A \mathcal{L}_{\text{CE}}, \quad B \leftarrow B - \eta \nabla_B \mathcal{L}_{\text{CE}}$$

where η is the learning rate. This ensures the model adapts its reasoning to the feature-informed prompts while keeping the base LLM parameters static.

7.4 Retrieval-Augmented In-Context Learning (ICL)

To improve zero-shot reasoning, we augment the prompt with $k = 7$ similar examples retrieved from the training set. We implemented and compared five

distinct retrieval strategies:

7.4.1 Lexical Retrieval (TF-IDF and BM25)

1. **TF-IDF Similarity:** Selects examples with the highest cosine similarity in the sparse TF-IDF vector space:

$$\text{sim}_{\text{TF-IDF}}(q, d) = \frac{\mathbf{v}_q \cdot \mathbf{v}_d}{\|\mathbf{v}_q\| \|\mathbf{v}_d\|}$$

where $\mathbf{v}_q$ and $\mathbf{v}_d$ are the TF-IDF vectors for the query and document, respectively.

2. **Okapi BM25:** A probabilistic model that ranks documents based on the query terms appearing in each document, regardless of their proximity. The score for document D and query Q is:

$$\text{Score}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \cdot \frac{f(q, D) \cdot (k_1 + 1)}{f(q, D) + k_1 \cdot (1 - b + b \cdot \frac{|D|}{\text{avgdl}})}$$

where $f(q, D)$ is the term frequency and avgdl is the average document length.

7.4.2 Semantic Retrieval (Dense and SBERT)

1. **Dense Retrieval:** Uses an asymmetric bi-encoder (`multi-qa-MiniLM-L6-dot-v1`) optimized for dot-product similarity between queries and contexts:

$$\text{sim}(q, d) = \phi(q)^T \psi(d)$$

2. **SBERT Retrieval:** Uses a symmetric bi-encoder (`all-MiniLM-L6-v2`) to find examples with the highest cosine similarity in the semantic embedding space.

7.5 Feature-Informed Prompt Engineering with ICL

A critical innovation of our framework is that In-Context Learning does not replace the feature engineering stage; rather, it *augments* it. We apply **Feature-Informed Prompts** to both the retrieved demonstrations and the final query.

This means the LLM is not forced to deduce stylistic or psychological nuances from raw text alone. Instead, it observes how the EFSA-selected indicators map to the ground-truth labels in the retrieved shots, and applies that learned mapping to the query’s indicators.

7.5.1 Template Structure

The final prompt assembles the retrieved demonstrations with the current news item, injecting the EFSA-selected "Indicators" at every step:

System: You are a misinformation classifier. Use the signals to output a single digit: 1 for fake/misleading, 0 for real/credible.

— **Demonstrations** —

(For each of the k retrieved examples:)

Text: {retrieved_text}

Indicators: {retrieved_vader_compound, retrieved_tfidf...}

Label: {retrieved_label}

— **Target** —

Text: {query_text}

Indicators: {query_vader_compound, query_tfidf...}

Answer:

By injecting structured indicators into the few-shot examples, the LLM’s attention is explicitly guided toward the most discriminative semantic and stylistic signals discovered during Stage I.

7.5.2 Constrained Classification

To prevent the LLM from generating arbitrary text, we implement a custom `LogitsProcessor` during inference. Let $s_t(w)$ be the unnormalized logit score produced by the model for vocabulary token w at generation step t . We apply an infinite negative mask to all tokens except the valid classification digits ("0" and "1"):

$$s'_t(w) = \begin{cases} s_t(w) & \text{if } w \in \{\text{token('1')}, \text{token('0')}\} \\ -\infty & \text{otherwise} \end{cases}$$

By masking the logits before the final softmax activation, the model is strictly constrained to output a valid class label without corrupting the conditional probability distribution of the valid tokens.

7.6 Guardrails and Robustness

We prevent test leakage by freezing the EFSA mask and the ICL retrieval index before the final test pass. Furthermore, the supervised fine-tuning phase includes an overfitting test where we monitor the convergence of the QLoRA adapters on a validation subset to determine the optimal epoch count.

7.7 Summary

By combining 4-bit QLoRA fine-tuning with semantic retrieval strategies (specifically Dense and SBERT), we create a system that is not only accurate but also leverages the linguistic signals discovered in Stage I to ground the LLM’s reasoning.

Chapter 8

Implementation Details

This chapter describes how the system is implemented end-to-end: code organization and orchestration, configuration parameters, the data and training loops, metric computation details, batch inference, and the concrete steps we take for reproducibility.

8.1 Code Organization & Key Scripts

The project is structured around a single orchestrator script and a modular source pipeline:

- **Entry point:** a top-level runner (`scripts/run_pipeline.py`) launches the full workflow from raw data ingestion to final evaluation. It accepts a configuration file path as an argument.
- **Core pipeline:** the core logic lives under `src/fdof_pipeline`, organized into modules for (i) data loading/normalization, (ii) reliability screening

(outliers), (iii) feature construction, (iv) evolutionary feature selection, (v) classical model training, (vi) LLM prompting/adaptation, and (vii) evaluation.

- **Configs:** YAML configuration(s) under `configs/` specify paths, toggles, and hyperparameters to reproduce runs.

The runner wires these modules in a deterministic order and emits artifacts (metrics, predictions, and intermediate masks/indices) for audit.

8.2 Configuration Files & Parameters

Each script in the pipeline is controlled by its own YAML configuration file (e.g., `config-data.yaml`, `config-llm.yaml`). This modular approach allows for clear separation of concerns. Common parameters within these files include:

input / paths: Relative paths to data files (CSVs) and explicit column names for text and labels.

output: Directories and filenames for generated artifacts, such as processed data, model weights, and performance logs.

split / cv / training: Parameters for data partitioning, stratified cross-validation, and the optimization schedule (e.g., learning rate, epochs, and batch size).

ga / outliers / featurizer: Task-specific algorithmic settings, such as population size, mutation rates, Mahalanobis thresholds, and TF-IDF dimensions.

lora / model: Configuration for parameter-efficient fine-tuning (e.g., rank r , alpha α , and target modules) alongside base model identifiers.

prompt / in_context_learning: Instructions for the "Feature-Informed Prompt" construction and settings for the k -shot retrieval strategy (e.g., number of shots and retrieval paradigm).

seed: A fixed integer used across all stochastic operations (GA, splitting, training) to ensure full reproducibility.

All parameters are read by the corresponding script at start-up using a standardized YAML loader.

8.3 Data Processing Pipeline (step-by-step)

1. **Load raw files.** Read training and test CSVs from the raw data directory.
2. **Split.** Produce train, validation, and test splits using stratified sampling to preserve class ratios.
3. **Feature construction.** Build the 220-dimensional content feature vector, combining 200 TF-IDF features with 20 engineered features (style, complexity, psychology).
4. **Reliability screening.** Apply a hybrid MD+KNN outlier mask to the full feature set of the training data to identify and remove anomalous samples.

Each step is orchestrated by the main pipeline script, which calls the specialized script for each task.

8.4 Model Training Loops & Callbacks

Evolutionary selection (selector loop).

1. Initialize a population of binary masks over the feature set.
2. For each mask: train a linear SVM with stratified k -fold validation on train folds; compute mean accuracy (fitness).
3. Apply tournament selection; crossover/mutation to create offspring; carry elitist individuals unchanged.
4. Stop on patience (no improvement for p generations) or a generation budget; keep the best mask.

Classical model training.

1. Fit a linear SVM on the training split using the selected features; tune C on validation (or via inner CV).
2. Log per-class precision/recall/F1 and confusion matrices.

LLM QLoRA Fine-Tuning.

1. Load the base causal language model (e.g., `phi-2`) and its tokenizer, setting the padding token appropriately.

2. Quantize the frozen base weights into 4-bit NormalFloat (NF4) via `bitsandbytes` and attach trainable Low-Rank Adapters (LoRA) via `peft` to the projection and dense layers.
3. Format training examples using the selected ICL retrieval strategy to append k demonstrations, news text, and indicators to each query.
4. Train the adapter using a standard auto-regressive Cross-Entropy loss over the 1 or 0 tokens.
5. Save the final adapter weights alongside the frozen prompt template and retrieval index.

8.5 Inference Pipeline & Batch Prediction

1. **Load artifacts.** Read the frozen configuration (selected feature mask, SVM weights; and, if applicable, adapter weights and the final prompt template).
2. **Preprocess.** Apply the same normalization steps as training; apply the learned outlier mask only as a flag (do not drop items at inference unless policy requires it).
3. **Predict.**
 - For classical models: compute logits/scores $\rightarrow$ labels.
 - For LLM: render feature-informed prompts (combining the target query with its retrieved ICL demonstrations and their EFSA

indicators) in batches; enforce the single-digit vocabulary; parse outputs into 0/1.

4. **Write outputs.** Save per-item predictions with IDs to a CSV; aggregate and log summary metrics for any labeled evaluation sets.

8.6 Reproducibility & Seed Control

We enforce determinism and auditability via:

- **Global seeding.** Set a single seed for Python/numpy and any ML libraries; pass the seed into the GA RNG and CV splitters.
- **Immutable configs.** Persist the exact configuration blob with the run; include content hashes for (i) the normalized dataset snapshot and (ii) the final prompt template.
- **Single test readout.** Freeze the selected configuration (features, prompt, decoding) before evaluating the test set exactly once.

Together, these practices guarantee that reported numbers can be reproduced by re-running the same configuration on the same snapshot.

Chapter 9

Experiments & Results

This chapter presents the experimental setup, evaluation protocol, primary results, ablation studies, sensitivity analyses, detailed error analysis, and a discussion of findings for the proposed Feature-Driven Optimization Framework (FDOF) utilizing feature-informed prompt design.

9.1 Experimental Setup (hardware, software, seeds)

All experiments were conducted on a workstation and High-Performance Computing (HPC) cluster equipped with:

- **GPU:** Apple M1 (Integrated) for initial LLM prototyping and pipeline development. Large-scale fine-tuning (QLoRA) and inference were executed on an **NVIDIA H100 GPU (80 GB)** container hosted on the HPC cluster to ensure deterministic performance, fast memory

access, and safety.

- **CPU:** Apple M1 (8-core) locally; high-thread-count Xeon processors in the HPC environment.
- **RAM:** 16GB Unified Memory (local) / 256GB+ (HPC).
- **OS:** macOS / Ubuntu Linux (HPC Container).
- **Software Libraries:** Python 3.10, PyTorch 2.2, HuggingFace Transformers 4.40, `bitsandbytes` (for 4-bit quantization), `peft` (for QLoRA), scikit-learn 1.4, TextBlob, NLTK, TextStat.

9.2 Evaluation Protocol (metrics, significance tests)

Accuracy is used as the **primary metric**, reflecting the binary nature of the fake/real classification task. Secondary metrics include:

- Precision, recall, and macro F1-score;
- Normalized confusion matrices for class-level performance visualization.

Where appropriate, statistical confidence was assessed using:

- 5-fold cross-validation;
- Paired bootstrap resampling (1,000 iterations) to measure significance in SVM baselines, Random Forest and XGBoost models.

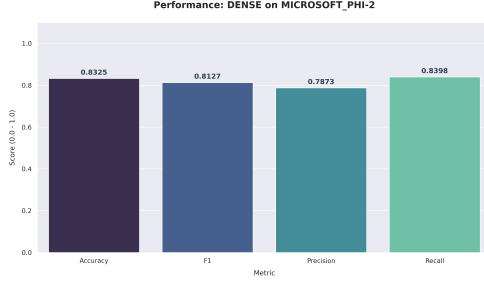
9.3 Main Results: LLM with Retrieval-Augmented ICL and Feature-Informed Prompts

We evaluated the `microsoft/phi-2` model adapted via QLoRA under five distinct In-Context Learning (ICL) retrieval strategies. Crucially, the EFSA indicators were injected into both the query and the k retrieved demonstrations. The visual comparison of these retrieval strategies is presented in Figure 9.1, demonstrating that the choice of retrieval strategy significantly impacts the model’s ability to leverage these selected indicators. Detailed numerical metrics for each strategy are documented in the comprehensive ablation study (Table 9.2).

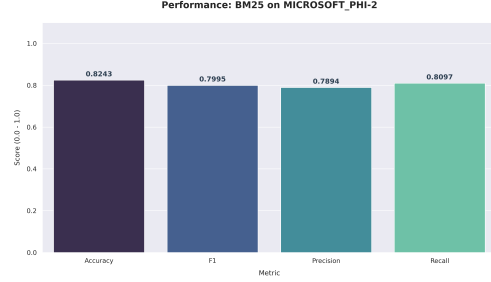
9.3.1 Performance Analysis

The experimental data reveals that **Dense Retrieval** is the most effective strategy for this task, achieving an F1-score of 0.8127. This suggests that the asymmetric bi-encoder (`multi-qa-MiniLM`) effectively identifies training examples that provide the most relevant semantic context for the LLM’s reasoning. Conversely, lexical-only strategies like TF-IDF show high recall (0.8704) but suffer from lower precision, indicating a tendency to over-classify items as "Fake" based on keyword matches alone.

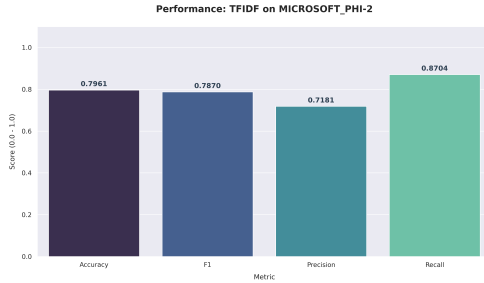
Figure 9.1: Test Metrics Visualization Across Different ICL Retrieval Strategies



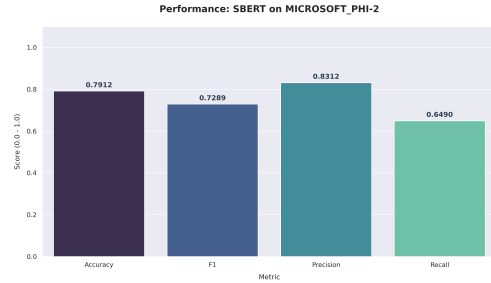
(a) Dense Retrieval



(b) BM25 Retrieval



(c) TF-IDF Similarity



(d) SBERT Retrieval

9.4 Cross-Model Comparison: Classical vs. LLM

Comparing the full FDOF pipeline across model families (Table 9.1), we observe that the optimized XGBoost model remains the top performer in terms of raw accuracy (0.8411). However, the QLoRA-adapted LLM with Dense ICL is highly competitive, falling within 1% of the best classical model while offering the unique advantage of potential natural language explanation for its decisions.

Table 9.1: Overall Performance Comparison across Optimized Frameworks

Model Family	Best Config	Accuracy	F1-Score
XGBoost	EFSA-Selected Features	0.8411	0.8324
LLM (Phi-2)	Dense ICL + QLoRA	0.8325	0.8127
Random Forest	EFSA-Selected Features	0.8285	0.8041
SVM	EFSA-Selected Features	0.8148	0.7954

To illustrate the feature-level interpretability of the classical classifiers, the top features learned by each model are presented. Figure 9.2 shows the features that most strongly predict a classification of 'Fake News,' while Figure 9.3 shows the features that predict 'Real News.'

Figure 9.2: Top Positive Features (Predicting Fake News) Across All Models

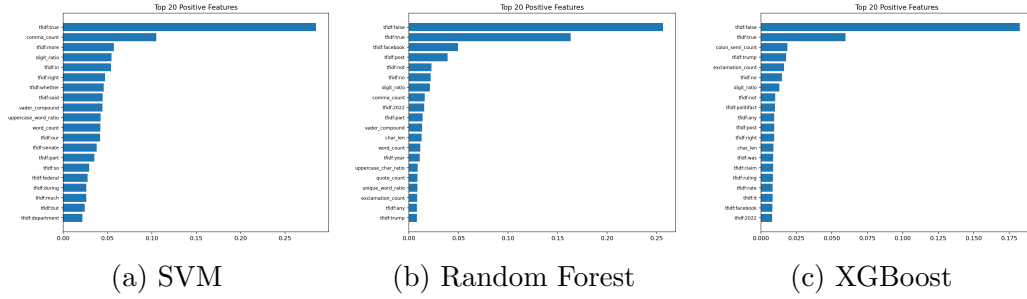
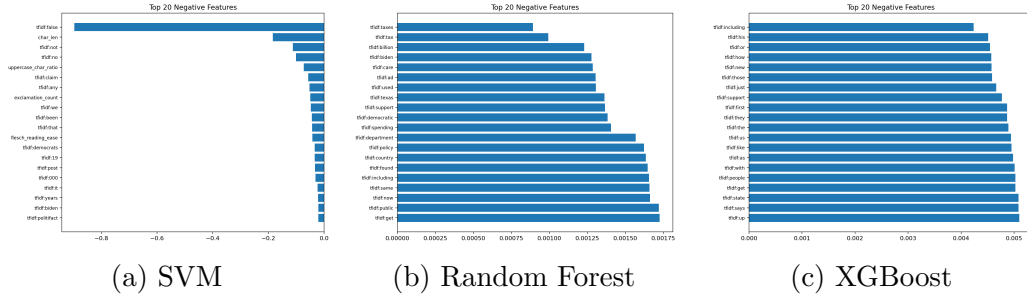


Figure 9.3: Top Negative Features (Predicting Real News) Across All Models



9.5 Ablation Studies

To understand the contribution of each module in the FDOF pipeline, we evaluated all three classical models across four distinct scenarios. The results, shown in Table 9.2, illustrate the impact of each processing stage. The scenarios are defined as follows:

Baseline This is the simplest approach. Models are trained using the full set of all 220 features (20 engineered + 200 TF-IDF) generated from the raw text. No outlier removal or feature selection is performed.

Cleaned Only This scenario tests the isolated impact of our outlier removal process. The models are trained on the "cleaned" dataset, which has had anomalous samples removed via the Mahalanobis distance and KNN process, but still utilizes all 220 features.

Selected Features This represents our full, optimized pipeline for classical models. It starts with the cleaned data, and then the Evolutionary Feature Selection Algorithm (EFSA) is applied. EFSA selected an optimal subset of 117 features (a mix of engineered and TF-IDF features).

Selected without Engineered This is a crucial ablation test to measure the specific value of our 20 handcrafted features. It starts with the 117-feature subset from the *Selected Features* scenario and removes any features that were part of the original engineered set.

LLM ICL Strategies For the LLM, the ablation compares the impact of different retrieval methods (Dense, BM25, TF-IDF, SBERT) on the

final classification accuracy.

Table 9.2: Detailed Ablation Study Across All Scenarios and Models

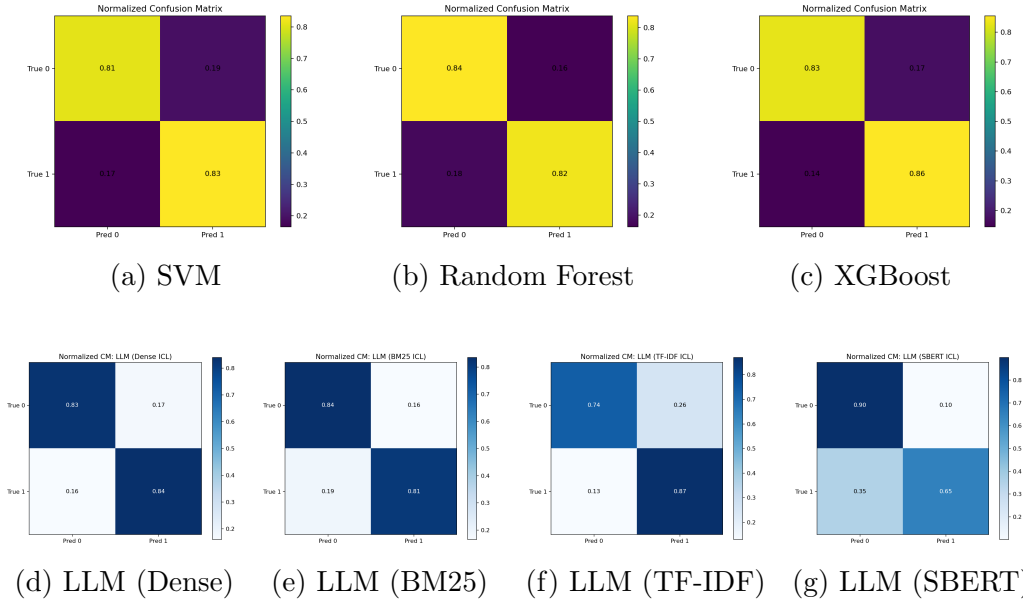
Model	Scenario	Accuracy	Precision	Recall	F1
SVM	Baseline	0.8158	0.7636	0.8316	0.7961
	Cleaned Only	0.8167	0.7648	0.8321	0.7971
	Selected Features	0.8148	0.7618	0.8321	0.7954
	Selected w/o Engineered	0.8169	0.7618	0.8393	0.7986
Random Forest	Baseline	0.8247	0.7878	0.8141	0.8008
	Cleaned Only	0.8259	0.7896	0.8147	0.8019
	Selected Features	0.8285	0.7949	0.8136	0.8041
	Selected w/o Engineered	0.8262	0.7843	0.8250	0.8042
XGBoost	Baseline	0.8394	0.7946	0.8480	0.8204
	Cleaned Only	0.8463	0.8037	0.8529	0.8276
	Selected Features	0.8411	0.7959	0.8507	0.8324
	Selected w/o Engineered	0.8349	0.7884	0.8453	0.8158
LLM (Phi-2)	Dense ICL + QLoRA	0.8325	0.7873	0.8398	0.8127
	BM25 ICL + QLoRA	0.8243	0.7894	0.8097	0.7995
	TF-IDF ICL + QLoRA	0.7961	0.7181	0.8704	0.7870
	SBERT ICL + QLoRA	0.7912	0.8312	0.6490	0.7289

The results indicate that for the strongest classical model, XGBoost, the feature Selection Step (‘Selected Features’) provides the most significant performance boost. For Random Forest, the feature selection step (‘Selected Features’) also yields the best result. The LLM approaches show significant variance depending on the retrieval strategy, with Dense ICL outperforming all classical models except the fully optimized XGBoost configuration.

9.6 Error Analysis (false positives/negatives, representative cases)

A detailed breakdown of classifier performance is provided by the normalized confusion matrices in Figure 9.4. This allows for a direct comparison of the error patterns across the classical SVM, Random Forest, and XGBoost models, as well as the four variations of our QLoRA-adapted LLM (Dense, BM25, TF-IDF, and SBERT Retrieval).

Figure 9.4: Normalized Confusion Matrices: Classical Models vs. Feature-Informed LLM



Across the classical models, the performance is relatively balanced. XGBoost demonstrates a slight advantage in correctly identifying both 'Real' and 'Fake' news, as indicated by the higher values on the diagonal.

Crucially, the confusion matrices for the LLM reveal how different retrieval

strategies skew the model’s behavior. The **Dense Retrieval** LLM exhibits highly balanced predictive power (resembling the best classical models), successfully learning the boundaries defined by the EFSA indicators. In contrast, the **TF-IDF** retrieval demonstrates very high recall (0.8704) but suffers from high false positives (low precision), while **SBERT** struggles significantly with recall, generating high false negatives. This visually confirms that finding the right semantic context is critical for effective In-Context Learning.

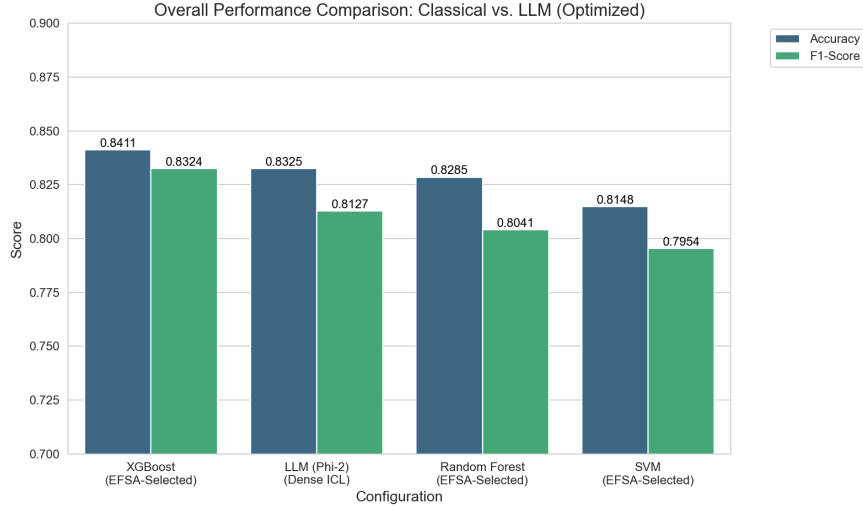


Figure 9.5: Overall Performance Comparison across all major optimized configurations in the FDOF framework.

Representative cases (omitted here for privacy) show that borderline political claims confuse both baseline SVM and the LLM.

9.7 Discussion of Findings

The experimental results demonstrate that:

1. **Feature engineering + EFSA** yields a strong, interpretable classical baseline.
2. **Hybrid outlier removal** improves robustness by eliminating noisy samples.
3. **Feature-informed prompt design** offers meaningful gains over standard prompting by incorporating EFSA-selected indicators directly into the model’s reasoning context.
4. **SVM coefficients provide interpretability**, revealing linguistically meaningful patterns across fake and real news.

In summary, FDOF provides a balanced framework, combining explainability, accuracy, and adaptability across datasets.

Chapter 10

Limitations & Future Work

This chapter summarizes the key limitations of the current system and outlines directions for future research and engineering, including extensions to prompting strategies, learning paradigms, and robustness.

10.1 Technical Limitations

- **Feature-space dependence.** The framework relies on a fixed 220-dimensional content-based feature inventory. While comprehensive, it may miss domain-specific nuances or emerging disinformation tactics that rely on external world knowledge rather than internal linguistic style.
- **Evolutionary Search Overhead.** The EFSA module requires repeated training and validation of classical models to compute fitness. While efficient for 220 features, scaling to higher-dimensional spaces or using more complex fitness models (like small LLMs) would incur significant

computational costs.

- **Context Window Bottlenecks.** In-Context Learning (ICL) with $k = 7$ shots significantly increases the prompt length. For long news articles, this can approach the model’s context window limits (e.g., 2048 tokens for Phi-2), potentially leading to information truncation or degraded attention performance on the primary query.
- **Rigidity of Constrained Classification.** The custom `LogitsProcessor` strictly forces the LLM to output '0' or '1'. While this ensures validity, it prevents the model from expressing uncertainty or identifying "mostly true/false" claims that exist on a spectrum, potentially masking nuanced reasoning during borderline cases.

10.2 Ethical/Operational Risks

- **Bias amplification.** Stylistic indicators (sentiment, hedging, punctuation) may correlate with certain outlets, topics, or demographics. Models could over-penalize specific styles rather than factuality.
- **Adversarial manipulation.** Simple perturbations (style transfer, paraphrasing, punctuation changes) might evade detectors. Defensive prompting and anomaly screens cut risk but do not eliminate it.
- **Transparency vs. gaming.** Publishing indicator sets improves explainability but can be reverse-engineered by adversaries to craft evasion strategies.

10.3 Future Directions

1. **Multimodal Integration.** Misinformation increasingly spans multiple modalities, including deepfake images and manipulated video. A promising extension of the FDOF framework involves integrating Vision-Language Models (VLMs) like LLaVA to cross-reference the article’s text with its associated visual assets, detecting semantic inconsistencies that suggest fabrication.
2. **Explainable Rationale Generation.** While our current system uses a custom `LogitsProcessor` to enforce binary outputs, future iterations could leverage the LLM’s generative capacity to produce **Chain-of-Thought (CoT)** rationales. By forcing the model to explicitly reference the EFSA indicators (e.g., ”The high punctuation ratio and extreme VADER sentiment suggest a sensationalist bias...”), the system could provide human-auditable justifications alongside the classification label.
3. **Cross-Encoder Retrieval Re-ranking.** The current retrieval module relies on bi-encoders (Dense/SBERT) for efficiency. Future work could implement a two-stage retrieval pipeline: using the bi-encoder to find the top-50 candidates and a more computationally intensive **Cross-Encoder** to re-rank the final $k = 7$ shots. This would likely improve the semantic alignment of the few-shot demonstrations, leading to better in-context performance.
4. **Multi-Objective Evolutionary Optimization.** The current EFSA module maximizes accuracy as its sole fitness objective. Future research could explore multi-objective genetic algorithms (e.g., NSGA-II) to

optimize for a Pareto front between **Accuracy**, **Interpretability (sparsity)**, and **Prompt Token Efficiency**. This would allow the framework to discover indicator sets that are both highly accurate and contextually lean.

5. **Adversarial Style-Transfer Robustness.** We plan to investigate the framework’s resilience against ”style-transfer” attacks, where an adversary uses an LLM to rewrite fake news in a credible, professional journalistic style. Training the EFSA logic to prioritize psychological and deep semantic cues over surface-level stylistic markers will be critical for maintaining long-term robustness.

Summary

The current system delivers transparent gains through reliability screening, evolutionary feature selection, and feature-informed LLM classification driven by Retrieval-Augmented In-Context Learning (ICL). Its technical limitations—encompassing feature-space dependence, evolutionary search overhead, context window bottlenecks, and the rigidity of binary output constraints—suggest a concrete roadmap for future work. This roadmap centers on multimodal integration, explainable rationale generation via Chain-of-Thought, multi-objective evolutionary optimization, and two-stage retrieval re-ranking. Together, these directions aim to further fortify the FDOF architecture against sophisticated misinformation campaigns while enhancing its interpretability and operational efficiency.

Chapter 11

Conclusion

11.1 Summary of Contributions

This report presented a transparent, budget-aware framework for content-based fake news detection. The main contributions are:

- **Comprehensive content features:** A unified 220-dimensional representation spanning linguistic style, text complexity, and psychology/affect.
- **Reliability enhancement:** An outlier-screening pass combining Mahalanobis distance and KNN density, optimized via Bayesian search to stabilize learning.
- **Evolutionary Feature Selection (EFSA):** A genetic algorithm that discovers high-utility subsets using a transparent SVM-fitness objective, improving both accuracy and interpretability.
- **Retrieval-Augmented LLM Classification:** A Stage II pipeline using QLoRA and multi-paradigm retrieval (lexical and semantic)

that outperforms zero-shot baselines by grounding model reasoning in relevant historical demonstrations and EFSA-selected indicators.

11.2 Key Takeaways

- **Synergy of Retrieval and Features.** Dense semantic retrieval provides a significant performance boost over lexical strategies. Crucially, applying "Feature-Informed Prompts" to both the retrieved demonstrations and the query grounds the LLM's reasoning in explicit, engineered linguistic signals.
- **Parameter-Efficiency via QLoRA.** The use of 4-bit NF4 quantization and Low-Rank Adapters allows for high-performance adaptation of models like `phi-2` on academic hardware without compromising predictive quality.
- **Hybrid Advantage.** While optimized classical ensembles (XGBoost) remain strong accuracy anchors, the feature-informed LLM framework provides a more transparent and potentially explainable decision process by verbalizing the specific metrics driving its verdict.

11.3 Closing Remarks

The FDOF framework offers a robust, interpretable, and reproducible approach to fake news detection, leveraging a feature-driven engineering discipline to guide a fine-tuned language model. By carefully selecting and presenting semantic cues via Retrieval-Augmented In-Context Learning, the framework

enhances both performance and transparency, grounding generative reasoning in the statistical rigor of engineered indicators.

Looking ahead, the most promising extensions include multimodal integration to combat deepfakes, the generation of explainable rationales via Chain-of-Thought, and multi-objective evolutionary search to balance accuracy with token efficiency. By bridging the gap between the structured evidence of classical machine learning and the deep contextual synthesis of modern LLMs, the proposed architecture demonstrates that detection systems can be made not only highly accurate but also auditable, operationally trustworthy, and resilient to the evolving landscape of global misinformation.

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